TECHNICAL UNIVERSITY OF MOLDOVA
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LABORATORY REPORT NO. 2

Cryptanalysis of Monoalphabetic Ciphers

Cryptography: frequency analysis and substitution recovery

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ABSTRACT

This laboratory examines the recovery of an English message encrypted with a monoalphabetic substitution. Letter frequencies provide initial hypotheses, while repeated words, common letter combinations and grammatical context constrain the remaining substitutions. The report reconstructs a 19-step analysis of Variant 1, supported by four supplied frequency screenshots. The cryptogram contains 2,567 letters and 25 distinct ciphertext symbols. All observed symbols are resolved consistently; the final unused alphabet pair is inferred by elimination. Intermediate extracts illustrate the gradual recovery of the passage, and the complete ciphertext and literal plaintext are preserved for comparison. The result demonstrates that a large substitution key space does not conceal the statistical structure of natural language.

Keywords: cryptanalysis, substitution, frequency analysis, word patterns, plaintext recovery

INTRODUCTION

The objective is to recover an English plaintext from a monoalphabetic cryptogram. The attack uses the frequency analysis method described in the laboratory handout [**?**]. Only letters are encrypted; numbers, spaces and punctuation provide additional information. The required report explains the successive deductions, as in the handout's worked example.

Variant 1 is used to match the supplied reference report by Gabriel Balan [**?**]. Its four frequency screenshots and deduction sequence are reused with attribution. The analysis below reconstructs the substitutions and verifies them against the Variant 1 ciphertext in the handout. Intermediate extracts are reconstructed text, not new captures of an interactive session. This laboratory requires a report only and contains no application implementation.

The report follows the structure of Laboratory Work No. 1: theoretical background, practical analysis, results and conclusions. Complete source and result texts are included in the appendices. The focus is on explaining why each proposed substitution is plausible and how it fits the rest of the message.

1 SHORT THEORY

A fixed substitution changes the labels of letters while preserving much of the structure that makes a language recognizable. The following properties form the basis of the attack described in the handout [**?**].

1.1 Monoalphabetic substitution

Let A be the 26-letter English alphabet and let π be a permutation of A. For plaintext letter p_i and ciphertext letter c_i, encryption and decryption are:

$$c_i = \pi(p_i), \qquad p_i = \pi^{-1}(c_i). \tag{1.1}$$

The same plaintext letter always produces the same ciphertext symbol. Consequently, letter counts, repeated-letter patterns and word lengths survive encryption. A general substitution has $26! \approx 4.03 \times 10^{26}$ possible keys, but this large key space does not prevent an attack based on language patterns. Unlike brute force, frequency analysis tries to infer the permutation from the message itself.

1.2 Letter frequencies

For a ciphertext of N alphabetic characters, the relative frequency of symbol c is:

$$f_C(c) = \frac{n_C(c)}{N} \times 100\%, \tag{1.2}$$

where $n_C(c)$ denotes its count. Uppercase and lowercase letters are counted together; digits, punctuation, spaces and line breaks are excluded. For a sufficiently long text, the distribution often resembles the average distribution of its language. English is used here because the assignment specifies an English plaintext.

Table 1.1 reproduces the English reference values from the handout. They support initial guesses, not a unique key: topic and vocabulary can substantially change a particular message's distribution.

Table 1.1 – English letter frequencies in percent, from the handout [?]

Letter	Frequency	Letter	Frequency
A	8.17	N	6.75
B	1.49	O	7.51
C	2.78	P	1.93
D	4.25	Q	0.09
E	12.70	R	5.99
F	2.23	S	6.33
G	2.01	T	9.06
H	6.09	U	2.76
I	6.97	V	0.98
J	0.15	W	2.36
K	0.77	X	0.15
L	4.03	Y	1.97
M	2.41	Z	0.07

1.3 Word patterns and consistency

Common English digraphs (two-letter sequences) include TH, HE, AN, IN, ER and ON. Common trigraphs (three-letter sequences) include THE, AND, THA, ENT and ION. The ordinary one-letter words *a* and *i*, doubled letters and repeated short words also help constrain the key. For example, the partial pattern `tQe` strongly suggests `the`.

A useful hypothesis must explain several occurrences without contradicting earlier deductions. Each ciphertext symbol has one plaintext value, and different ciphertext symbols have different plaintext values. A guessed letter that produces one readable word but breaks several others should be reconsidered. Proper names and rare words become useful later, after common words have established most of the key.

2 PRACTICAL ANALYSIS

The practical work reconstructs the solution sequence in the supplied reference [**?**]. All mappings below run from ciphertext to plaintext: $V \rightarrow e$ means that ciphertext V represents plaintext e. In partial decryptions, unresolved symbols are uppercase and resolved letters are lowercase.

2.1 Frequency evidence

The supplied cryptogram contains 2,567 letters and uses 25 distinct ciphertext symbols. Ciphertext M does not occur. Figures 2.1 and 2.2 show the ranked frequency tables extracted from the reference report. The integer counts in the ciphertext screenshot agree with the counts obtained from Variant 1 in the handout.

V	W	N	X	G	T	I	P	Q	O	H	S	Z	F	C	U	D	J	A	R	K	L	Y	B	E	M
327	276	205	198	196	178	176	165	145	93	83	80	63	61	59	58	56	45	35	27	17	13	8	2	1	0
12.7	10.8	8.0	7.7	7.6	6.9	6.9	6.4	5.6	3.6	3.2	3.1	2.5	2.4	2.3	2.3	2.2	1.8	1.4	1.1	0.7	0.5	0.3	0.1	0.0	0.0

Figure 2.1 – Ranked ciphertext frequencies, reused from the reference report [?]

E	T	A	O	I	N	S	H	R	D	L	C	U	M	W	F	G	Y	P	B	V	K	J	X	Q	Z
12.7	9.1	8.2	7.5	7.0	6.7	6.3	6.1	6.0	4.3	4.0	2.8	2.8	2.4	2.4	2.2	2.0	2.0	1.9	1.5	1.0	0.8	0.15	0.15	0.10	0.07

Figure 2.2 – Ranked English frequencies, reused from the reference report [?]

Ciphertext V appears 327 times, giving $327/2567 \times 100\% \approx 12.74\%$. Its frequency is close to the English reference of 12.70% for e. Ciphertext W appears 276 times, approximately 10.75%, and is a candidate for t. The screenshot rounds these values to 12.7% and 10.8%.

Ranking alone is insufficient. Ciphertext N is third with 205 occurrences, but represents o, whereas T is sixth with 178 occurrences and represents a. These substitutions must be established from words, rather than by matching two sorted lists mechanically.

The alphabetically ordered profiles in Figures 2.3 and 2.4 illustrate the same statistical clues. The strongest English peaks occur at E and T; the strongest ciphertext peaks occur at V and W. These are separate distributions: the passage need not match average English exactly.

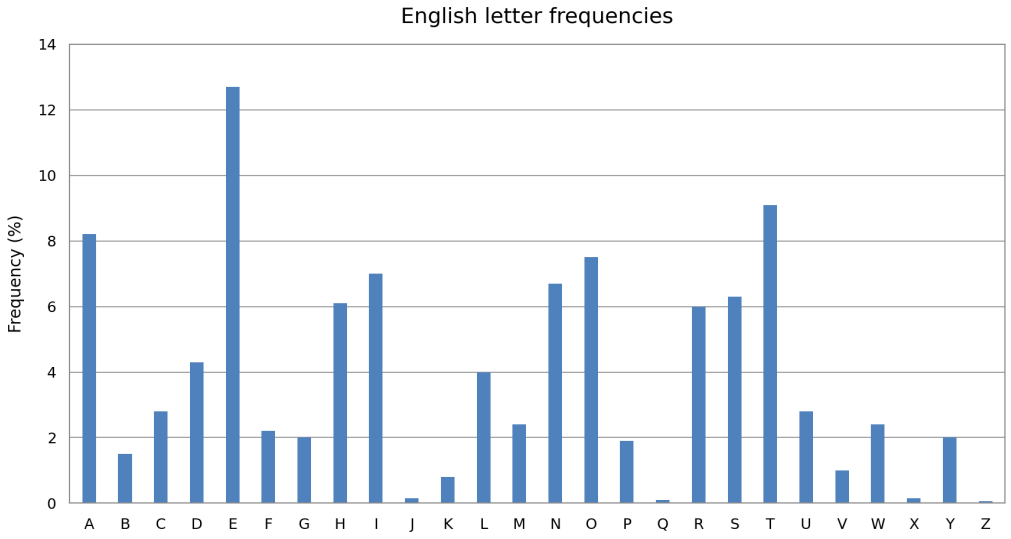


Figure 2.3 – English frequency profile, reused from the reference report [?]

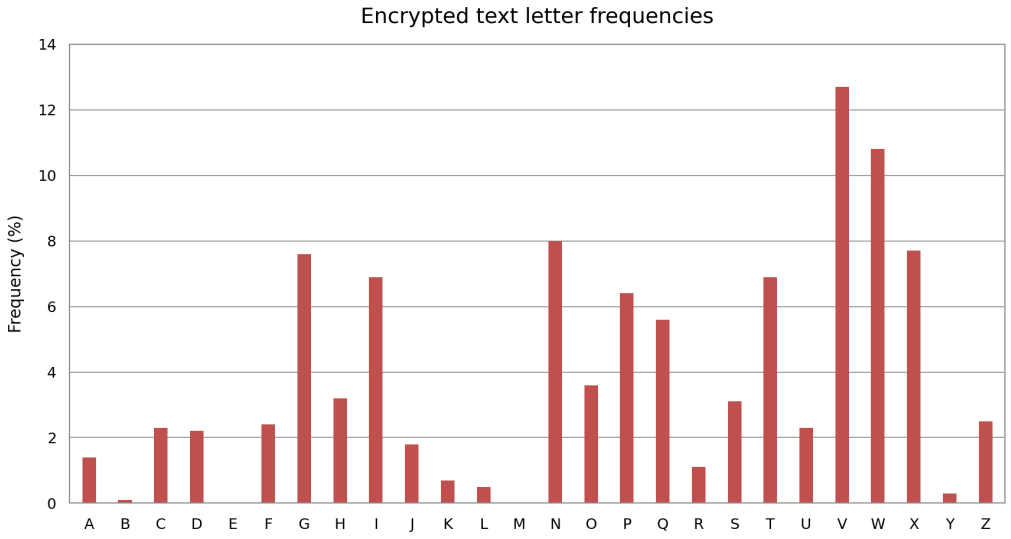


Figure 2.4 – Ciphertext frequency profile, reused from the reference report [?]

2.2 Steps 1–7: common letters

The first substitutions combine frequency hypotheses with short recurring words. The sequence follows the reference report [**?**]:

1. select $V \to e$: V occurs 327 times (12.74%), making it the strongest candidate for the most frequent English letter;
2. select $W \to t$: W occurs 276 times (10.75%); this second major peak suggests t, and the two replacements expose repeated instances of `tQe`;
3. infer $Q \to h$: interpreting `tQe` as `the` confirms a common word and supports the previous substitutions;
4. infer $T \to a$: `thTt` becomes `that`, and the same symbol also appears as a one-letter word;
5. infer $P \to s$: `thePe` becomes `these`; P occurs 165 times (6.43%), close to the English reference for s;
6. infer $N \to o$: the frequent two-letter pattern `tN` becomes `to`; its 205 occurrences (7.99%) are consistent with a common vowel;
7. infer $X \to i$: `Xt` becomes `it`, and 198 occurrences (7.71%) support a common letter;

2.3 Steps 8–12: phrases

Once short words are readable, longer phrases constrain several unknown letters at once:

8. infer $Z \to m$, $I \to r$ and $G \to n$: `the saZe IeasoG as` becomes `the same reason as`; the phrase supplies three compatible mappings;
9. infer $O \to d$: `Oesire to reaO` becomes `desire to read`, with the same letter fitting both ends of the phrase;
10. infer $F \to y$, $S \to l$ and $J \to g$: `on a daF nearSF 4,000 Fears aJo` becomes `on a day nearly 4,000 years ago`; repeated F receives the same value in three words;
11. infer $R \to w$ and $H \to c$: `in a toRn Halled menet` becomes `in a town called menet`; the ordinary words can be solved without first identifying the place name;
12. infer $C \to f$: `story oC his lord's liCe` becomes `story of his lord's life`, resolving both a common preposition and a noun;

2.4 Steps 13–19: remaining observed letters

Context supplies the less frequent letters that are difficult to identify statistically:

13. infer $L \to k$: `world Lnows it` becomes `world knows it`;
14. infer $D \to u$ and $A \to b$: `a master scriAe sketched oDt` becomes `a master scribe sketched out`; these pairs also clarify `tomb` and `about`;
15. infer $U \to p$: `thehieroglyUhs` becomes `thehieroglyphs`; the joined words reflect a missing space in the supplied text;
16. infer $B \to q$: `techniBue` becomes `technique`; B appears only twice, making word context more informative than its frequency;
17. infer $K \to v$: `eKen` becomes `even` in the phrase "even the slightest"; this pair also resolves `developed` and `service`;
18. infer $Y \to x$: `read the teYt` becomes `read the text`; Y occurs eight times;
19. infer $E \to j$: `instead of Eust writing` becomes `instead of just writing`; E occurs once and is the last observed ciphertext symbol resolved;

3 RESULTS

The successive substitutions recover a coherent passage about early Egyptian writing and cryptology. The following extracts make the accumulation of information visible before the complete key and plaintext are presented.

3.1 Progressive decryption

Each extract uses the same opening of the ciphertext. The case convention distinguishes unknown symbols from recovered letters. The excerpts are reconstructed from the cumulative substitutions above, with line wrapping adjusted for the report. Original missing spaces are retained.

3.1.1 After steps 1 and 2

```
NG T OTF GeTISF 4,000 FeTIP TJN, XG T tNRG HTSSeO ZeGet LQDCDANIOeIXGJ tQe tQXG
IXAANG NC tQe GXSe, T ZTPteI PHIXAe PLetHQeO NDt tQeQXeINJSFUQP tQTt tNSO tQe PtNIF
NC QXP SNIO'P —SXCe TGO XG PN ONXGJ QeNUeGeO tQe IeHNIOeO QXPtNIF NC HIFUtNSNJF. QXP
RTP GNt T PFPteZ NC
```

3.1.2 After step 3

```
NG T OTF GeTISF 4,000 FeTIP TJN, XG T tNRG HTSSeO ZeGet LhDCDANIOeIXGJ the thXG
IXAANG NC the GXSe, T ZTPteI PHIXAe PLetHheO NDt thehXeINJSFUhP thTt tNSO the PtNIF
NC hXP SNIO'P —SXCe TGO XG PN ONXGJ heNUeGeO the IeHNIOeO hXPtNIF NC HIFUtNSNJF. hXP
RTP GNt T PFPteZ NC
```

3.1.3 After step 4

```
NG a OaF GeaISF 4,000 FeaIP aJN, XG a tNRG HaSSeO ZeGet LhDCDANIOeIXGJ the thXG
IXAANG NC the GXSe, a ZaPteI PHIXAe PLetHheO NDt thehXeINJSFUhP that tNSO the PtNIF
NC hXP SNIO'P —SXCe aGO XG PN ONXGJ heNUeGeO the IeHNIOeO hXPtNIF NC HIFUtNSNJF. hXP
RaP GNt a PFPteZ NC
```

3.1.4 After steps 5 and 6

```
oG a OaF GeaISF 4,000 FeaIs aJo, XG a toRG HaSSeO ZeGet LhDCDAoIOeIXGJ the thXG
IXAAoG oC the GXSe, a ZasteI sHIXAe sLetHheO oDt thehXeIoJSFUhs that toSO the stoIF
oC hXs SoIO's —SXCe aGO XG so OoXGJ heoUeGeO the IeHoIOeO hXstoIF oC HIFUtoSoJF. hXs
Ras Got a sFsteZ oC
```

3.1.5 After steps 7 and 8

```
on a OaF nearSF 4,000 Fears aJo, in a toRn HaSSeO menet LhDCDAorOerinJ the thin
riAAon oC the niSe, a master sHriAe sLetHheO oDt thehieroJSFUhs that toSO the storF
oC his SorO's —SiCe anO in so OoinJ heoUeneO the reHorOeO historF oC HrFUtoSoJF. his
Ras not a sFstem oC
```

3.1.6 After steps 9 and 10

```
on a day nearly 4,000 years ago, in a toRn Halled menet LhDCDAordering the thin
riAAon oC the nile, a master sHriAe sLetHhed oDt thehieroglyUhs that told the story
oC his lord's —liCe and in so doing heoUened the reHorded history oC HryUtology. his
Ras not a system oC
```

3.1.7 After steps 11 and 12

```
on a day nearly 4,000 years ago, in a town called menet LhDfDAordering the thin
riAAon of the nile, a master scriAe sLetched oDt thehieroglyUhs that told the story
of his lord's —life and in so doing heoUened the recorded history of cryUtology. his
was not a system of
```

3.1.8 After steps 13–15

All letters in this opening extract are now resolved; the remaining four observed symbols occur elsewhere in the passage.

```
on a day nearly 4,000 years ago, in a town called menet khufubordering the thin
ribbon of the nile, a master scribe sketched out thehieroglyphs that told the story
of his lord's —life and in so doing heopened the recorded history of cryptology. his
was not a system of
```

3.1.9 After steps 16–19

The opening is unchanged. The final deductions resolve the words `technique`, `even`, `text` and `just` elsewhere.

```
on a day nearly 4,000 years ago, in a town called menet khufubordering the thin
ribbon of the nile, a master scribe sketched out thehieroglyphs that told the story
of his lord's —life and in so doing heopened the recorded history of cryptology. his
was not a system of
```

3.2 Recovered substitution alphabet

Table 3.1 gives the complete decryption alphabet. The 19 steps identify 25 observed pairs because several steps establish more than one letter. Every observed ciphertext letter has a distinct plaintext value.

Table 3.1 – Recovered ciphertext-to-plaintext substitution

Ciphertext	A	B	C	D	E	F	G	H	I	J	K	L	M
Plaintext	b	q	f	u	j	y	n	c	r	g	v	k	z*
Ciphertext	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Plaintext	o	d	s	h	w	l	a	p	e	t	i	x	m

*M does not occur; $M \rightarrow z$ is inferred by elimination.

Under the full-alphabet permutation assumption, the sole unused plaintext letter z must correspond to the sole unused ciphertext letter M. No occurrence in the passage directly confirms this last pair.

3.3 Verification

The recovered key was applied to every letter of Variant 1, preserving nonletter characters. Applying the inverse key to the resulting literal plaintext reproduces the ciphertext after case normalization, including its punctuation, whitespace and line breaks. This confirms consistent application of the mapping. A round trip alone cannot establish the intended plaintext; grammatical English, repeated word patterns and the independent contextual deductions provide the additional evidence.

The ciphertext contains 2,567 alphabetic characters. Its 26 symbol counts agree with the supplied frequency screenshot, including the zero count for M. The literal plaintext is preserved in Appendix B; the ciphertext appears in Appendix A. Missing spaces such as `thehieroglyphs` are present in the source and do not represent a failed substitution.

3.4 Complete plaintext: reading version

The following passage is recovered from the supplied cryptogram [**?**, **?**]. For readability, obvious word boundaries and capitalization are restored, and the stray bullet before “proliferated” is omitted. These editorial changes do not alter the sequence of letters. The historical claims belong to the decrypted passage. On a day nearly 4,000 years ago, in a town called Menet Khufu bordering the thin ribbon of the Nile, a master scribe sketched out the hieroglyphs that told the story of his lord’s life—and in so doing he opened the recorded history of cryptology. His was not a system of secret writing as the modern world knows it; he used no fully developed code of hieroglyphic symbol substitutions. His inscription, carved about 1900 B.C. into the living rock in the main chamber of the tomb of the nobleman Khnumhotep II, merely uses some unusual hieroglyphic symbols here and there in place of the more ordinary ones. Most occur in the last 20 columns of the inscription’s 222, in a section recording the monuments that Khnumhotep had erected in the

service of the pharaoh Amenemhet II.

The intention was not to make it hard to read the text. It was to impart a dignity and authority to it, perhaps in the same way that a government proclamation will spell out “In the year of Our Lord One thousand eight hundred and sixty three” instead of just writing “1863.” The anonymous scribe may also have been demonstrating his knowledge for posterity. Thus the inscription was not secret writing, but it incorporated one of the essential elements of cryptography: a deliberate transformation of the writing. It is the oldest text known to do so.

The transformations occur in funerary formulas, in a hymn to Thoth, in a chapter of the Book of the Dead, on the sarcophagus of the pharaoh Seti I, in royal titles displayed in Luxor, on the architrave of the Temple of Luxor, on stele, in laudatory biographic inscriptions. There is nothing meant to be concealed in all this; indeed, many of the statements are repeated in ordinary form right next to the altered ones. Why, then, the transformations? Sometimes for essentially the same reason as in Khnumhotep’s tomb: to impress the reader. Occasionally for a calligraphic or decorative effect; rarely, to indicate a contemporary pronunciation; perhaps even for a deliberate archaism as a reaction against foreign influence.

But many inscriptions are tinctured, for the first time, with the second essential for cryptology—secrecy. In a few cases, the secrecy was intended to increase the mystery and hence the arcane magical powers of certain religious texts. But the secrecy in many more cases resulted from the understandable desire of the Egyptians to have passersby read their epitaphs and so confer upon the departed the blessings written therein.

In Egypt, with its concentration upon the afterlife, the number of these inscriptions soon proliferated to such an extent that the attention and the goodwill of visitors flagged. To revive their interest, the scribes deliberately made the inscriptions a bit obscure. They introduced the cryptographic signs to catch the reader’s eye, make him wonder, and tempt him into unriddling them—and so into reading the blessings. It was a sort of Madison Avenue technique in the Valley of the Kings. But the technique failed utterly. Instead of interesting the readers, it evidently destroyed even the slightest desire to read the epitaphs, for soon after the funerary cryptography was begun, it was abandoned.

CONCLUSIONS

The analysis recovers all 25 ciphertext symbols present in the message through 19 deduction steps. Frequencies suggest the initial e and t substitutions, but word patterns and grammatical context establish the remaining pairs. The unused $M \rightarrow z$ pair follows only from elimination under the permutation assumption.

The exercise demonstrates why frequency ranking is a starting point rather than a complete solution. A readable short excerpt is also insufficient: symbols occurring only later must still be resolved. Checking the complete passage confirms that each recovered letter has one consistent interpretation.

The result connects directly with Laboratory Work No. 1. Enlarging the available alphabet permutations does not conceal repeated letters or the frequency distribution when substitution remains fixed throughout the message. The weakness exploited here is the preservation of language structure, even when exhaustive key search is impractical.

REPORT MATERIALS

The materials are stored under `lab2`. They include the report source, four reused screenshots, ciphertext and literal plaintext. The laboratory repository is available at `https://github.com/yoda-dinmd/crypt-labs`.

APPENDICES

APPENDIX A

COMPLETE SUPPLIED CIPHERTEXT

The Variant 1 text below preserves the letters, punctuation, missing spaces and line breaks extracted from the supplied handout [**?**]. Display lines may wrap to fit the page; case is ignored during analysis.

```
NG T OTF gvtisf 4,000 fvtip tjn, xg t wnrg htssvo Zvgvw Lqdcdaniovixgj wqv wqxg
    ixaang nc wqv
Gxsv, t ztpwvi phixav plvwhqvo ndw wqvqxvinjsfuqp wqtw wnso wqv pwnif nc qxp snio'p —
    sxcv
tgo xg pn onxgj qvnuvgvo wqv ivhniovo qxpwnif nc hifuwnsnjf. Qxp rtp gnw t pfpwvz nc
pvhivwrixwxgj tp wqv znovig rniso lgnrp xw; qv dpvo gn cdssf ovkvsnuvo hnov nc
qxvinjsfuqxhpfzans pdapwxwdwxngp. Qxp xgphixuwxng, htikvo tandw 1900 A.H. xgwn
    wqvsxkxgj
inhl xg wqv ztxg hqtzavi nc wqv wnza nc wqv gnasvztgLqgdzqnwvu XX, zvivsf dpvp pnzv
    dgdpdts
qxvinjsfuqxh pfzansp qvivtgo wqviv xg usthv nc wqv zniv nioxgtif ngvp. Znpw nhhdi xg
    wqv stpw
20hnsdzgp nc wqv xgphixuwxng'p 222, xg t pvhwxng ivhnioxgj wqv zngdzvgwpwqtw
    Lqgdzqnwvu
qto vivhwvo xg wqv pvikxhv nc wqv uqtitnq TzvgvzqvwXX. Wqv xgwvgwxng rtp gnw wn ztlv
    xw
qtio wn ivto wqv wvyw. Xw rtp wnxzutiw t oxjgxwf tgo tdwqnixwf wn xw, uviqtup xg wqv
    ptzv rtf
wqtw tjnkvigzvgw uinhstztwxng rxss puvss ndw "Xg wqv fvti nc Ndi Snio Ngvwqndptgo
    vxjqw
qdgoivo tgo pxywf wqivv" xgpwvto nc edpw rixwxgj "1863."Wqv tgngfzndp phixav ztf tspn
     qtkv
avvg ovzngpwitwxgj qxp lgnrsvojvcni unpwvixwf. Wqdp wqv xgphixuwxng rtp gnw pvhivw
rixwxgj, adw xwxghniunitwvo ngv nc wqv vppvgwxts vsvzvgwp nc hifuwnjituqf: t
ovsxavitwvwitgpcniztwxng nc wqv rixwxgj. Xw xp wqv nsovpw wvyw lgnrg wn on pn.Wqv
witgpcniztwxngp nhhdi xg cdgvitif cnizdstp, xg t qfzg wn Wqnwq,xg t hqtuwvi nc wqv
    Annl nc wqv
Ovto, ng wqv ptihnuqtjdp nc wqv uqtitnqPvwx X, xg infts wxwsvp oxpustfvo xg Sdyni, ng
     wqv
tihqxwitkv nc wqv Wvzusv nc Sdyni, ng pwvsv, xgstdotwnif axnjituqxh xgphixuwxngp.
    Wqviv xp
gnwqxgj zvtgw wn avhnghvtsvo xg tss wqxp; xgovvo, ztgf nc wqv pwtwvzvgwp tiv ivuvtwvo
xgnioxgtif cniz ixjqw gvyw wn wqv tswvivo ngvp. Rqf, wqvg, wqvwitgpcniztwxngp?
    Pnzvwxzvp
cni vppvgwxtssf wqv ptzv ivtpng tp xgLqgdzqnwvu'p wnza: wn xzuivpp wqv ivtovi.
    Nhhtpxngtssf
cni thtssxjituqxh ni ovhnitwxkv vccvhw; itivsf, wn xgoxhtwv t
    hngwvzunitifuingdghxtwxng; uviqtup
vkvg cni t ovsxavitwv tihqtxpz tp t ivthwxngtjtxgpw cnivxjg xgcsdvghv.Adw ztgf
    xgphixuwxngp tiv
wxghwdivo, cni wqv cxipw wxzv, rxwq wqvpvhngo vppvgwxts cni —hifuwnsnjfpvhivhf. Xg t
    cvr
htpvp, wqv pvhivhf rtpxgwvgovo wn xghivtpv wqv zfpwvif tgo qvghv wqv tihtgv ztjxhts
    unrvip
nchviwtxg ivsxjxndp wvywp. Adw wqv pvhivhf xg ztgf zniv htpvp ivpdswvo cinzwqv
dgovipwtgotasv ovpxiv nc wqv Vjfuwxtgp wn qtkv utppvipaf ivto wqvxivuxwtuqp tgo pn
    hngcvi
```

dung wqv ovutiwvo wqv asvppxgjp rixwwvg wqvivxg.Xg Vjfuw, rxwq xwp hnghvgwitwxng dung wqv tcwvisxcv, wqv gdzavi nc wqvpvxgphixuwxngp pnng • uinsxcvitwvo wn pdhq tg vywvgw wqtw
wqv twwvgwxng tgowqv jnnorxss nc kxpxwnip cstjjvo. Wn ivkxkv wqvxi xgwvivpw, wqv phixavpovsxavitwvsf ztov wqv xgphixuwxngp t axw naphdiv. Wqvf xgwinodhvo wqvhifuwnjituqxh
pxjgp wn htwhq wqv ivtovi'p vfv, ztlv qxz rngovi, tgowvzuw qxz xgwn dgixoosxgj wqvz — tgo pn
xgwn ivtoxgj wqv asvppxgjp. Xwrtp t pniw nc Ztoxpng Tkvgdv wvhqgxbdv xg wqv Ktssvf nc wqv
Lxgjp. Adwwqv wvhqgxbdv ctxsvo dwwvisf. Xgpwvto nc xgwvivpwxgj wqv ivtovip, xwvkxovgwsf
ovpwinfvo vkvg wqv psxjqwvpw ovpxiv wn ivto wqv vuxwtuqp, cnipnng tcwvi wqv cdgvitif hifuwnjituqf rtp avjdg, xw rtp tatgongvo.

APPENDIX B

LITERAL DECRYPTED TEXT

The recovered key is applied to every alphabetic character, with plaintext written in lowercase. All nonletter characters and original word boundaries are retained. Unlike the reading version, joined words and the source bullet remain visible.

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on a day nearly 4,000 years ago, in a town called menet khufubordering the thin
    ribbon of the
nile, a master scribe sketched out thehieroglyphs that told the story of his lord's —
    life
and in so doing heopened the recorded history of cryptology. his was not a system of
secretwriting as the modern world knows it; he used no fully developed code of
hieroglyphicsymbol substitutions. his inscription, carved about 1900 b.c. into
    theliving
rock in the main chamber of the tomb of the noblemankhnumhotep ii, merely uses some
    unusual
hieroglyphic symbols hereand there in place of the more ordinary ones. most occur in
    the last
20columns of the inscription's 222, in a section recording the monumentsthat
    khnumhotep
had erected in the service of the pharaoh amenemhetii. the intention was not to make
    it
hard to read the text. it was toimpart a dignity and authority to it, perhaps in the
    same way
that agovernment proclamation will spell out "in the year of our lord onethousand
    eight
hundred and sixty three" instead of just writing "1863."the anonymous scribe may also
     have
been demonstrating his knowledgefor posterity. thus the inscription was not secret
writing, but itincorporated one of the essential elements of cryptography: a
deliberatetransformation of the writing. it is the oldest text known to do so.the
transformations occur in funerary formulas, in a hymn to thoth,in a chapter of the
    book of the
dead, on the sarcophagus of the pharaohseti i, in royal titles displayed in luxor, on
     the
architrave of the temple of luxor, on stele, inlaudatory biographic inscriptions.
    there is
nothing meant to beconcealed in all this; indeed, many of the statements are repeated
inordinary form right next to the altered ones. why, then, thetransformations?
    sometimes
for essentially the same reason as inkhnumhotep's tomb: to impress the reader.
    occasionally
for acalligraphic or decorative effect; rarely, to indicate a
    contemporarypronunciation; perhaps
even for a deliberate archaism as a reactionagainst foreign influence.but many
    inscriptions are
tinctured, for the first time, with thesecond essential for —cryptologysecrecy. in a
    few
cases, the secrecy wasintended to increase the mystery and hence the arcane magical
    powers
ofcertain religious texts. but the secrecy in many more cases resulted fromthe
understandable desire of the egyptians to have passersby read theirepitaphs and so
    confer
upon the departed the blessings written therein.in egypt, with its concentration upon
```

the afterlife, the number of theseinscriptions soon • proliferated to such an extent that
the attention andthe goodwill of visitors flagged. to revive their interest, the scribesdeliberately made the inscriptions a bit obscure. they introduced thecryptographic
signs to catch the reader's eye, make him wonder, andtempt him into unriddling them — and so
into reading the blessings. itwas a sort of madison avenue technique in the valley of the
kings. butthe technique failed utterly. instead of interesting the readers, itevidently
destroyed even the slightest desire to read the epitaphs, forsoon after the funerary cryptography was begun, it was abandoned.